A 2-ADIC OBSTRUCTION TO MODULAR LYAPUNOV FUNCTIONS FOR THE $3n+1$ PROBLEM

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ABSTRACT. Let T be the accelerated Collatz map: $T(n) = n/2$ for even n, $T(n) = (3n+1)/2$ for odd n. We prove, by an elementary and self-contained argument, that for every $j \geq 1$ no function of the form $V(n) = \log_2 n + w(n \bmod 2^j)$, with $w : \mathbb{Z}/2^j\mathbb{Z} \to \mathbb{R}$ arbitrary, decreases strictly at every positive odd integer. The obstruction comes from the fixed point $-1 \in \mathbb{Z}_2$: its projection at every finite level carries positive representatives with logarithmic gain $\log_2 3 - 1 > 0$, while the modular correction cancels. The conclusion persists with finite exceptions, non-strict inequality, and blocks of k iterations. The hypothesis is imposed only on $\mathbb{Z}_+$, where no cycle other than $\{1, 2\}$ is known. **Question:** does this positive-integer formulation appear explicitly in the literature?

1. Statement and proof

Let $T : \mathbb{Z} \to \mathbb{Z}$ be the accelerated Collatz map, $T(n) = n/2$ for n even and $T(n) = (3n + 1)/2$ for n odd. The Collatz conjecture asserts that every $n \geq 1$ reaches the cycle $\{1, 2\}$ under iteration of T [2, 3]. Since the length-k parity vector is determined by the residue modulo 2^k [5], finite periodic corrections to the logarithmic scale form a natural class of candidate descent certificates. The result below shows that this class is empty.

Definition 1 (Modular Lyapunov function)**.** Fix $j \geq 1$. A *modular Lyapunov function of level j* is a function

$$V(n) \;=\; \log_2 n + w(n \bmod 2^j), \qquad w : \mathbb{Z}/2^j\mathbb{Z} \to \mathbb{R} \text{ arbitrary},$$

such that $V(T(n)) < V(n)$ for every odd integer $n > 0$.

Since w has finite domain, it is bounded and acts on the integers as a periodic potential of period 2^j. If such a V existed, no positive orbit could diverge.

Theorem 2 (Modular obstruction)**.** *For every $j \geq 1$ and every $w : \mathbb{Z}/2^j\mathbb{Z} \to \mathbb{R}$, the function $V(n) = \log_2 n + w(n \bmod 2^j)$ is not strictly decreasing along the odd steps of T in $\mathbb{Z}_+$. Specifically, the projection of the 2-adic fixed point $-1 \in \mathbb{Z}_2$ onto $\mathbb{Z}/2^j\mathbb{Z}$ induces, on the class $-1 \bmod 2^j$, a loop of logarithmic gain $\log_2 3 - 1 > 0$ that no potential w compensates.*

Proof. Suppose, for contradiction, that there is $w : \mathbb{Z}/2^j\mathbb{Z} \to \mathbb{R}$ such that, for every odd $n > 0$,

$$w\big(T(n) \bmod 2^j\big) - w\big(n \bmod 2^j\big) \;<\; -\log_2 \frac{T(n)}{n}. \tag{1}$$

Consider the family $n_m = 2^{j+1}m - 1$, $m \geq 1$. Since $j \geq 1$ we have $n_m \geq 3$; every n_m is therefore a strictly positive odd integer, contained in the domain where (1) is assumed. The accelerated step evaluates exactly:

$$T(n_m) \;=\; \frac{3(2^{j+1}m - 1) + 1}{2} \;=\; 3 \cdot 2^j m - 1.$$

Reducing modulo 2^j,

$$n_m = 2^j(2m) - 1 \equiv -1, \qquad T(n_m) = 2^j(3m) - 1 \equiv -1 \pmod{2^j}.$$

Substituting n_m into (1), the potential terms cancel identically, leaving, for every $m \geq 1$,

$$0 \;<\; -\log_2 \frac{3 \cdot 2^j m - 1}{2 \cdot 2^j m - 1}.$$

This requires the argument of the logarithm to be strictly less than 1; since numerator and denominator are positive, that means $3 \cdot 2^j m - 1 < 2 \cdot 2^j m - 1$, i.e. $3 < 2$ — a contradiction. $\square$

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Remark 3 (The mechanism). In $\mathbb{Z}_2$ the integer -1 is an odd fixed point of the accelerated map: $T(-1) = (3\cdot(-1)+1)/2 = -1$. Any function of the residues modulo 2^j is blind to the difference between the positive integers $n_m \equiv -1 \pmod{2^{j+1}}$, which we want to control, and the point $-1 \in \mathbb{Z}_2$ they approach 2-adically, on which the dynamics genuinely gains $\log_2 \frac{3}{2} = \log_2 3 - 1 > 0$ per step. The family n_m transports this stationary gain into $\mathbb{Z}_+$, where the potential cancels and only the gain remains. The obstruction is therefore *structural*: it does not depend on the combinatorics of w, only on the existence of the 2-adic fixed point.

2. Robustness

Corollary 4 (Robustness)**.** *The obstruction of Theorem 2 persists under each of the following relaxations of Definition 1.*

(1) Finite exceptions. *If* (1) *is only required outside a finite set $E \subset \mathbb{Z}_+$, no such V exists.*
(2) Non-strict decrease. *If the requirement is weakened to $V(T(n)) \le V(n)$, no such V exists.*
(3) Blocks of k iterations. *If the requirement is $V(T^k(n)) < V(n)$ for a fixed $k \ge 1$, no such V exists.*
(4) Restricted domains. *If the decrease is required only on a subset $D \subseteq \mathbb{Z}_+$ of the odd integers, then either D omits infinitely many terms of some family $n_m = 2^{j+1}m - 1$ — in which case V cannot certify global convergence — or D contains infinitely many of them, and no such V exists.*

Proof. (1) The family $\{n_m\}_{m\ge 1}$ is infinite; for any finite E there is m with $n_m \notin E$, and the proof of Theorem 2 applies verbatim to that m.

(2) With $\le$ in place of $<$, the final inequality becomes $3 \le 2$, still false.

(3) Parametrize the deeper family $n_m = 2^{j+k}m - 1$, $m \ge 1$. By induction, $T^i(n_m) = 3^i\, 2^{j+k-i}m - 1$ for $0 \le i \le k$, each of the first k steps being odd: if $i < k$, then $j + k - i \ge j + 1 \ge 2$, so $3^i 2^{j+k-i}m$ is even, $T^i(n_m)$ is odd, and

$$T^{i+1}(n_m) = \frac{3\bigl(3^i 2^{j+k-i}m - 1\bigr) + 1}{2} = 3^{i+1}2^{j+k-i-1}m - 1.$$

Hence $T^k(n_m) = 3^k 2^j m - 1 \equiv -1 \equiv n_m \pmod{2^j}$: the potentials cancel in the block inequality, leaving $0 < -\log_2 \frac{3^k 2^j m - 1}{2^k 2^j m - 1}$, which forces $3^k < 2^k$, false for every $k \ge 1$.

(4) If D contains infinitely many n_m, run the proof of Theorem 2 on those; if it omits all but finitely many, the orbits through the omitted integers are left entirely unconstrained by V. □

3. Relation to the literature, and the question

That *some* obstruction to modular Lyapunov functions must exist over all of $\mathbb{Z}$ is unsurprising: the negative integers contain genuine cycles (-1; $-5 \to -7 \to -10$; and the cycle of -17), and a strictly decreasing V on $\mathbb{Z} \setminus \{0\}$ would forbid them. Moreover, the underlying phenomenon is folklore: it has been understood since the stopping-time analyses of Terras [5] and Everett [1] that the residue class $-1 \bmod 2^k$ rises for k consecutive accelerated steps (e.g. $n = 2^k - 1$), so no descent argument reading only a fixed finite dyadic residue can be uniform.

The content of Theorem 2 is the precise packaging of this phenomenon as a *nonexistence* statement for the class of Definition 1: the hypothesis is imposed *only on* $\mathbb{Z}_+$, where no cycle other than $\{1, 2\}$ is known, and the failure is nevertheless unconditional — caused by the projection of the single 2-adic point -1, not by any cycle in the domain of the hypothesis. We make no claim of priority; we regard the theorem as a formalization of a known heuristic, in the spirit of the average-descent results of [5, 4].

The question motivating this note: is this formulation — nonexistence of a strictly decreasing $V(n) = \log_2 n + w(n \bmod 2^j)$ *on the positive odd integers*, with the robustness of Corollary 4 — stated explicitly anywhere in the $3x + 1$ literature? References or corrections are most welcome.

The full manuscript (with additional Wasserstein-contraction results and a reproducible exact-arithmetic computational laboratory) is available at `https://github.com/tuliomarchetto/collatz`.

References

[1] C. J. Everett, *Iteration of the number-theoretic function $f(2n) = n$, $f(2n+1) = 3n+2$*, Advances in Mathematics **25** (1977), 42–45.
[2] J. C. Lagarias, *The $3x+1$ problem and its generalizations*, American Mathematical Monthly **92** (1985), 3–23.
[3] J. C. Lagarias (ed.), *The Ultimate Challenge: The $3x+1$ Problem*, American Mathematical Society, Providence, RI, 2010.
[4] T. Tao, *Almost all orbits of the Collatz map attain almost bounded values*, Forum of Mathematics, Pi **10** (2022), e12.
[5] R. Terras, *A stopping time problem on the positive integers*, Acta Arithmetica **30** (1976), 241–252.

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